

Supporting Information for

Large language models put existential risk from artificial intelligence at 10%

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Supporting Information Text

This text extends the Materials and Methods of the main text. It gives the full prompt (section 1), the selection of models and settings (2), the data collection (3), the variables (4), the statistical methods (5) and alternative model specifications (6). Model outputs are in Datasets S1–S3: the models and their inclusion decisions (S1), the answers per model and reasoning setting (S2) and all model estimates (S3). Raw replies, including the explanations, and the analysis code are on GitHub (https://github.com/Teyfice/llm-existential-risk-estimates).

1. Prompt

Each call sent the following text as a single user message. The question is the general-framing question of the 2023 Expert Survey on Progress in AI, word for word (1, p. 9:43). The four causes, with their definitions and examples, are taken in shortened form from Hendrycks et al. (2, pp. 1–2). Their order was randomized for each call (the same order in the list and in the JSON template).

> What probability do you put on future AI advances causing human extinction or similarly permanent and severe disempowerment of the human species?
>
> Answer with:
>
> 1. probability: Your best estimate, as a probability between 0 and 1. Use as many decimal places as you need.
>
> 2. lower_bound and upper_bound: Probabilities between 0 and 1 such that you are 90% confident the probability lies between them.
>
> 3. cause: If this outcome occurred, what would most likely be its primary cause? Choose exactly ONE:
>
> - `"malicious_use"`: Individuals or groups intentionally harnessing powerful AIs to cause widespread harm, e.g., bioterrorism enabled by AIs that can help humans create deadly pathogens; the deliberate dissemination of uncontrolled AI agents; or the use of AI capabilities for propaganda, censorship, and surveillance.
> - `"ai_race"`: Competitive environments compelling actors to deploy unsafe AIs or cede control to AIs, e.g., militaries developing autonomous weapons, enabling automated warfare where accidents can spiral out of control; corporations automating human labor and prioritizing profits over safety, potentially leading to dependence on AI systems; or evolutionary pressures among AIs that could eventually lead to the displacement of humanity.
> - `"organizational_risks"`: Human factors and complex systems increasing the chances of catastrophic accidents, e.g., organizations developing advanced AIs suffering catastrophic accidents, particularly if they do not have a strong safety culture; AIs being accidentally leaked to the public or stolen by malicious actors; or organizations suppressing internal concerns about AI risks.
> - `"rogue_ai"`: The inherent difficulty in controlling agents far more intelligent than humans, e.g., AIs optimizing flawed objectives to an extreme degree (proxy gaming); goal drift as AIs adapt to a changing environment; AIs becoming power-seeking; or deception, with AIs appearing to be under control when they are not.
>
> These causes can interact. Choose the one you consider the most important driver.
>
> 4. explanation: A brief justification of your judgment (max. 150 words).
>
> Respond ONLY with a valid JSON object in exactly this format, with no other text:

```
{
  "probability":  <number between 0 and 1>,
  "lower_bound":  <number between 0 and 1>,
  "upper_bound":  <number between 0 and 1>,
  "cause":  "<malicious_use | ai_race | organizational_risks | rogue_ai>",
  "explanation":  "<string>"
}
```

2. Models

Catalogue. The OpenRouter catalogue downloaded on 5 November 2026 listed 452 entries. 108 were removed automatically as duplicates or non-models: batch versions (73), aliases (19), routers (9) and other duplicates (7). This left 344 models.

Inclusion criteria and decisions. A model was included if its OpenRouter description presents it as a general-purpose language model for chat, following instructions or reasoning that answers from the model itself; also handling images, video, audio or code did not disqualify it. Each of the 344 descriptions was coded by hand. 249 models were included and 95 excluded, for coding (25), roleplay or creative writing (14), image generation (10), narrow perceptual or control tasks (8), systems combining several models (6), answers through live web search (5), single document tasks (4), blocked by the account's privacy setting (4), aliases of listed models (4), faster serving variants of listed models (4), safety classification (3), translation (3), music output (2), speech output (2) and a single domain (1). Dataset S1 gives every decision with its category and reason.

Reasoning settings. Each model was queried at every reasoning setting it offers on OpenRouter. For the 241 analyzed models these were the listed effort levels (102 models), reasoning switched on and off (70), a single setting for models with mandatory reasoning (12) or a single setting for models without reasoning control (57). If reasoning was mandatory, the effort "none" was not requested. For the analysis, settings were mapped to seven reasoning levels: none/off (effort "none", reasoning off, or no reasoning control), minimal, low, medium/on (effort "medium", reasoning on, or mandatory reasoning), high, xhigh and max. The 649 analyzed cells (model × setting) were distributed over the levels as follows: none/off (184), minimal (20), low (90), medium/on (161), high (98), xhigh (51) and max (45). 168 models answered at two or more levels; 73 at one.

Default setting. Descriptive comparisons (main text Fig. 1*A* and *E*) use one value per model at its default setting as listed by OpenRouter: the default effort (102 models), the single setting of models with mandatory reasoning or without reasoning control (12 and 57), and for on/off models the state OpenRouter enables by default (on for 23, off for 4). For the 43 on/off models without a listed default, the value is the mean over both settings. The shares of causes (main text Fig. 1*E*) are each model's share of runs naming a cause at its default setting, averaged over the models of a region.

3. Data collection

Requests. All models were queried through the OpenRouter API (chat completions endpoint, https://openrouter.ai/api/v1). Each call sent the prompt as one user message, with no system prompt, no web access and no sampling parameters, so the provider's defaults applied. The reasoning setting was sent as a request parameter, and OpenRouter was asked to use only providers supporting all parameters sent. The output limit was 32,768 tokens or, if lower, the provider's listed limit (51 model–setting combinations) or the context length minus 1,000 tokens (2). OpenRouter routed each call to one of the providers hosting the model; providers were not selected by us. The valid answers of 181 of the 241 models came from a single provider (64 providers in total). The five runs of all 666 model–setting combinations were started on 6 November 2026 (3,330 calls, about US$35 in total). A pilot run with an earlier version of the prompt was not analyzed.

Answers. An answer was valid if the reply contained a JSON object with the estimate and both bounds as numbers between 0 and 1, one of the four causes and an explanation. Of the 3,330 calls, 3,220 gave a valid answer (96.7%); 419 of these embedded the JSON object in other text. The others were technical failures (70), replies without usable JSON (36) and JSON with missing or invalid values (4).

Analyzed models. Of the 249 included models, 6 gave no valid answer in any run, and the developers of 2 could not be assigned to a region. This left 241 models from 33 developers, with 3,210 valid answers in 649 cells; 627 cells had a valid answer in all five runs and 22 in one to four.

4. Variables

Developer region. Developers were assigned to the region where they are based: US (17 developers), China (13) and other (3). Developer names are the prefixes of the OpenRouter model identifiers (meta-llama merged with meta).

Release date. The release date is the date a model was first listed on OpenRouter. Release dates ranged from 28 May 2023 to 4 November 2026 and entered the models in years relative to 1 January 2025.

Context length, price and open weights. Context length (tokens) and input price (US dollars per million input tokens) are the values listed by OpenRouter, both on the $\log_{10}$ scale. The 2 free models without a paid version were set to $0.001, the lowest value on the axis. A model has open weights if OpenRouter lists a Hugging Face repository for it (119 models).

AI researchers. Researcher answers are the cleaned, anonymized responses of the 2023 Expert Survey on Progress in AI (1) (https://osf.io/vmdny) to the same question (variable `extinction_basic_clean`, in percent; divided by 100). 1,321 respondents answered it.

5. Statistical analysis

Beta mixed models. Answers were averaged over the valid runs of each cell. No cell mean was exactly 0 or 1, so the means entered the models unchanged. The estimate, upper bound and lower bound (models M1–M3) were each modeled as

$$y_i \sim \mathrm{Beta}\big(\mu_i\phi,\,(1-\mu_i)\phi\big), \qquad \mathrm{logit}(\mu_i) = \mathbf{x}_i^{\top}\boldsymbol{\beta} + u_{d(i)} + v_{m(i)},$$

with mean μ_i, precision ϕ (constant; variance $\mu_i(1-\mu_i)/(1+\phi)$), random intercepts $u_d \sim N(0,\sigma_u^2)$ for developers and $v_m \sim N(0,\sigma_v^2)$ for models within developers (3). Fixed effects $\mathbf{x}_i$ were region (reference: US), release date (per year), $\log_{10}$ context length, $\log_{10}$ input price (per tenfold increase), open weights and reasoning level (six indicators, reference: medium/on). Exponentiated coefficients are odds ratios (ORs) with Wald 95% CIs. Models were fitted by maximum likelihood with glmmTMB 1.1.13 (4) in R 4.5.3 (5); all converged with a positive-definite Hessian.

Tests. Region and reasoning level were tested jointly with Wald tests of their two and six indicators. The joint test of region gave $P = 0.069$ for the estimate, $P = 0.005$ for the upper bound and $P = 0.16$ for the lower bound.

Reasoning-level differences. Differences between reasoning levels (main text Fig. 1*D*) are average marginal differences in the 168 models that answered at two or more levels: every cell of these models was set to a level, with each model's random intercepts kept, and the predicted values were averaged and compared with those at medium/on. 95% CIs are percentiles over 4,000 draws from the multivariate normal distribution of the fixed effects.

Typical model. The lines in main text Fig. 1*B* and *C* are model-implied values for a typical model: random intercepts at 0 and the other covariates, including the region and reasoning-level indicators, at their means over cells. Release date runs from the 2% quantile of the models (June 2024) to the latest release (November 2026), input price from its 2% to its 98% quantile ($0.03 to $16). Bands are 95% CIs, computed on the logit scale from the fixed-effect covariance and transformed to probabilities.

Temporal convergence. Whether newer models lie closer together was tested on the default-setting values, one per model. In a beta mixed model of these values with release date in the mean, a random intercept for developers and log precision depending on release date, exp of the release-date coefficient of the precision is the precision ratio per year (2.43, 95% CI 1.84–3.21, $P < 0.001$). A ratio above 1 means that the values of newer models scatter less around their mean. The standard deviations reported in the main text compare the 119 models released before the median release date (5 March 2026) with the 122 released from then on.

Comparison with AI researchers. The spread of the 241 default-setting values and the 1,321 researcher answers was compared with the Brown–Forsythe test (deviations from the group medians) (6). Means in main text Fig. 1*A* have *t*-based 95% CIs.

Reliability across runs. For the 648 analyzed cells with at least two valid runs, the one-way random-effects intraclass correlation (ICC) (7) was estimated with the ANOVA estimator for unequal numbers of runs. ICC(1) is the reliability of a single run's answer; ICC(k) is that of the mean of $k = 5$ runs (Spearman–Brown), the values analyzed. 95% CIs are percentiles over 4,000 bootstrap resamples of models, keeping all cells of a drawn model.

References to human judgments. An explanation was counted as referring to human judgments if it contained one of the words survey(s), surveyed, poll(s), polled, researcher(s), expert(s), scholar(s), forecaster(s) or superforecaster(s) (case-insensitive, whole words). This applied to 18% of the 3,210 valid answers of the analyzed models.

6. Alternative model specifications

Reasoning level as a linear step. Entering the reasoning level as one linear step (0 for none/off to 6 for max), split into the deviation from each model's mean (within models) and the model mean (between models) (8) (model A1), gave per step within models ORs of 1.00 for the estimate (95% CI 0.99–1.01, $P = 0.96$), 1.03 for the upper bound (1.02–1.04, $P < 0.001$) and 0.96 for the lower bound (0.95–0.98, $P < 0.001$). The between-model parts were 0.98 (0.90–1.07, $P = 0.61$), 1.03 (0.94–1.11, $P = 0.56$) and 0.88 (0.79–0.97, $P = 0.011$). Likelihood-ratio tests against within-model level indicators (model A2) did not reject linearity for the estimate ($P = 0.18$) and the upper bound ($P = 0.82$), but did for the lower bound ($P = 0.017$).

Reasoning tokens instead of reasoning level. The reasoning level correlated strongly with the number of reasoning tokens reported for an answer ($\log_{10}(\text{tokens} + 1)$, mean over runs; $r = 0.86$ within models, $r = 0.79$ across all cells). With this measure instead of the reasoning-level indicators (model A3), the ORs per tenfold increase were 0.99 for the estimate (95% CI 0.98–1.00, $P = 0.19$), 1.04 for the upper bound (1.02–1.06, $P < 0.001$) and 0.93 for the lower bound (0.91–0.95, $P < 0.001$); the other effects were similar.

SI Dataset S1 (Dataset_S1_models.csv)

Models. One row for each of the 344 models left after the automatic exclusions: developer; inclusion decision, category, reason and close-call flag; whether analyzed and, if not, why; region and its source; release date (OpenRouter listing); context length; input price and its source; open weights; reasoning settings queried and the default setting; and the answer at the default setting (estimate, bounds and share of runs naming each cause).

SI Dataset S2 (Dataset_S2_answers.csv)

Answers. One row for each model and reasoning setting queried: reasoning level; whether it is the default setting and part of the analysis; number of calls and of valid answers in the five runs; mean estimate, bounds and reasoning tokens over the valid answers; and the number of valid answers naming each cause.

SI Dataset S3 (Dataset_S3_model_estimates.xlsx)

Model estimates. One sheet each for the fixed effects of all models (M1–M3 and alternative specifications A1–A3, with odds ratios and Wald 95% CIs), the joint tests, the random-effect variances, model fit, the reasoning-level differences, the typical-model values, the temporal convergence model and the reliability measures. A first sheet describes each.

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